Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100305

Subject: Digital Electronics

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) What is the 2's complement of binary number 0101?
(i) 1010
(ii) 1111
(iii) 1011
(iv) 0110

(b) In a 4-variable K-map, how many cells are there?
(i) 4
(ii) 8
(iii) 12
(iv) 16

(c) A ring counter is a type of:
(i) Shift register with feedback
(ii) Memory unit
(iii) Synchronous counter
(iv) Asynchronous counter

(d) The resolution of a 10-bit A/D converter is:
(i) 10 V
(ii) 1/1024 of full-scale input
(iii) 1/100 of full-scale input
(iv) Depends on input frequency

(e) Content Addressable Memory (CAM) is best suited for:
(i) Fast search applications
(ii) Sequential data access
(iii) Arithmetic processing
(iv) Address-based data retrieval

(f) Which A/D converter uses a binary search algorithm?
(i) Counting type
(ii) Dual-slope type
(iii) Successive approximation
(iv) Flash type

(g) Which of the following is a characteristic of ROM?
(i) Volatile
(ii) Read/Write
(iii) Used as cache memory
(iv) Non-volatile and Read-only

(h) Which gate is known as the "universal gate"?
(i) OR
(ii) NAND
(iii) XOR
(iv) AND

(i) How many select lines are needed for a 16:1 multiplexer?
(i) 2
(ii) 8
(iii) 4
(iv) 16

(j) Which of the following devices is a 1-bit memory element?
(i) Flip-flop
(ii) AND gate
(iii) Decoder
(iv) Encoder

Q.2 (a) What are error-detecting and error-correcting codes? Discuss parity bits and Hamming code in detail? [7]

(b) Compare the characteristics of TTL, Schottky TTL, and CMOS logic families. [7]

Q.3	(a)	Compare and explain the operation of a half adder, full adder, and carry look-ahead adder.	[7]
	(b)	Design a 2:1 multiplexer using basic gates. Extend it to a 4:1 multiplexer and explain its working with a truth table.	[7]
Q.4	(a)	Differentiate between synchronous and asynchronous counters. Design a 3-bit asynchronous ripple counter and explain its operation.	[7]
	(b)	Explain the working of a bistable latch using NOR gates. What are its properties, and how is it used in memory elements?	[7]
Q.5	(a)	Explain the working principle of the weighted resistor D/A converter.	[7]
	(b)	Describe the process of quantization and encoding in Analog-to-Digital conversion. Explain how quantization error affects the accuracy of A/D conversion.	[7]
Q.6	(a)	Describe the structure and operation of a Programmable Logic Array (PLA). How does it differ from a Programmable Array Logic (PAL)?	[7]
	(b)	Describe the operation of Read-Only Memory (ROM) and its various types such as PROM, EPROM, and EEPROM. What are their advantages and limitations?	[7]
Q.7	(a)	Compare the architecture and functioning of RAM and CAM. Highlight their respective use cases.	[7]
	(b)	What is the function of a sample-and-hold circuit in data acquisition systems? Explain its internal operation with the help of a suitable diagram.	[7]
Q.8	(a)	Explain the operation of basic digital logic gates: AND, OR, NOT, NAND, NOR, and XOR with their truth tables. Also, give one practical application for each.	[7]
	(b)	What is Tri-state logic? Explain its significance in bus systems?	[7]
Q.9	(a)	What is the Quine–McCluskey (Q-M) method? Apply it to minimize a 3-variable Boolean function: $F(A,B,C) = \Sigma(1,2,3,5,7)$.	[7]
	(b)	Explain the working of a sequence generator using flip-flops. Design a sequence generator for the sequence 1010.	[7]